

Camunda DISP Testing

Test name	Expected result	Result of the test	Goal achieved
Test 1: Select product	The form takes the product category, product name, quantity and if it's for hire or sale	The form took the input fields and submitted accordingly	Yes
Test 2: Use tool confirmation	The form takes the tool name, expected return date and time, tool condition	The form took the input fields and submitted accordingly	Yes
Test 3: Calculate the price	The price is given by the database and it multiplied by quantity and it returns a total	There is no database, but a default price is multiplied by quantity	Partially
Test 4: Check the availability	The process checks a database for the availability of an item	There is no database	No
Test 5: Delivery, click and collect or direct purchase	User is required to select an option for collecting their purchase, and only needs to give an address for delivery	User can select an option, but has to give an address no matter what, and collection time field can be filled with non time formats	Partially

Test 6: select payment method	The form takes payment method selection(card,finance and cash but credit and card number is only required if card is chosen, trade card number is only required if finance is chosen)	Credit card number is required no matter what choice of payment is chosen	Partially
Test 7: process payment	Payment is successful	Payment is successful	Yes
Test 8: Validate if user is a fintrust member	User is rejected if they do not give a trade card number or put 0 as the number	User is sometimes rejected no matter what, sometimes accepted	No
Test 9: Fintrust pays for the customer	Payment is done by Fintrust if user is a valid trade card member	Payment is done by Fintrust if user is a valid trade card member	Yes
Test 10: Customer repayment	Simulate repayment by calculating the required monthly payment	Repayment is calculated by dividing total price by 12	Yes
Test 11: Stock level update	Java code gets the product name and quantity and updates stock in the database	Java code gets product name and quantity, but there is no database. Updated stock level is simulated instead	Partially
Test 12: Create shipment	Shipment ID is created, and shipment info is logged	Shipment ID is created, and shipment info is logged. Only happens if the user has selected delivery	Yes
Test 13: Send delivery notification	Simulated notification by logging customer info, shipment ID and the product	Simulated notification by logging customer info, shipment ID and the product, but only the customer ID, shipment ID and product name is logged. No delivery address, collection time or product category is logged	Partially

Test 14: Use tool form	All fields are required. Agreeing to tools and conditions, date and time, and tool condition. Product name is prefilled and cannot be changed	All fields are required, product name is prefilled but can be changed	Partially
Test 15: Categorise tool:	User inputs whether the tool needs routine maintenance, repairs or is completely out of service	User inputs whether the tool needs routine maintenance, repairs or is completely out of service	Yes
Test 16: Decommission tool	Log info of the tool along with product name and customer ID	Log info of the tool along with product name and customer ID	Yes
Test 17: Generate report	Generate reports including product information, and customer ID, simulated by logging them	Product name and customer ID are logged, but not quantity or product category	Partially

Conclusion

For the most part, the entire business process is accurately simulated, however there are some areas that can be improved.

The biggest improvement that could be made would be the addition of a database. This could store product information and some trade card numbers that could be used for validation. The “Check Availability” service task could look for the input product name in the database, and return whether it is available or not depending on whether it is in the database, and if so, whether the quantity requested by the customer is more than or equal to the quantity in the database. Stock levels can then update these quantities, rather than simulating it. Additionally, the FinTrust Pool can check the database for valid trade card numbers, then validate whether the user is a valid member based on that.

Another improvement that could be made would be extra additions to the forms, such as some fields only being required if specific options are chosen. For example, if the customer selects click and collect or direct purchase as the collection option, they should not be forced to input their address. However, if they do, the address field should be validated by checking for the correct format, rather than just validating based on input length. Additionally, the collection time input should be a date and time format, rather than a text field. This way, only a date and time could be input, rather than the process accepting any input.